

Comparative Analysis of Naive Bayes, Decision Tree and XG-Boost for Crop Recommendation System

Ayushi Butani

Department of Computer Engineering
Vidyavardhini's College of Engineering and
Technology
Mumbai, India

Dinesh Lakshmanan

Department of Computer Engineering
Vidyavardhini's College of Engineering and
Technology
Mumbai, India

Nidhi Mehta

Department of Computer Engineering
Vidyavardhini's College of Engineering and
Technology
Mumbai, India

Abstract. *The need for increasing crop production is a significant challenge for farmers. This is due to a combination of factors including climate change, soil degradation, and limited access to resources and technology. All of these factors can lead to decreased yields and, in turn, reduced income for farmers. To meet this challenge, farmers must develop new methods of production that are resilient, cost effective, and sustainable. This will require a holistic approach, involving technological and agricultural advances, as well as improved access to resources, investments, and training. Crop cultivation anywhere in the world depends on the climate, seasons and various factors like soil properties. However, the process of increasing the production of crops depends on various factors, including mainly on temperature. Our proposed method utilises Machine Learning procedures with the end goal that it proposes appropriate crops dependent on the temperature, humidity, potassium, nitrogen, rainfall, pH, and phosphorus. This kind of framework helps to reduce the losses faced by farmers brought about by the probability of bad harvests, and furthermore gives information on the seasonal characterization of yields and which harvest is reasonable for which season. There are several machine learning algorithms available in this system, including KNN, Decision Tree, Random Forest, Naive Bayes and XG-Boost.*

Index Terms. Naive Bayes, Decision Tree, XG-Boost, Machine Learning, Crop Recommendation System.

I. INTRODUCTION

There is no doubt that the major source of income in rural India is agriculture and its allied industries. Moreover, agriculture contributes a lot to the country's GDP. The country is fortunate to have such a large agricultural sector. However, this may be one of the possible reasons for the higher suicide rate among marginal farmers in India. For farmers, this study offers a practical and easy to use yield forecasting system.

The proposed system provides connectivity to farmers through mobile or online applications. User location can be determined using GPS. The user enters a location and a soil type. Machine learning algorithms help select the most profitable crop lists or forecasts. Some machine learning

techniques are used to predict crop yields, including Support Vector Machines (SVM), Decision Trees (DT), XGBoost, Gaussian Naive Bayes, Random Forest (RF), and K Nearest Neighbors (KNN). Among them, Random Forest showed strong results with around 90% accuracy, while Naive Bayes showed around 99%.

Agriculture in India is vital to the socio economic structure of the country. For a country where nearly 58% of the population works in agriculture, the inability of farmers to use traditional, unscientific methods to choose the crops best suited to the soil is a serious problem. Farmers sometimes choose the wrong crops based on soil properties, planting season and location. As a result, some farmers commit suicide, stop farming and move to cities in search of work.

This research work proposes a mechanism to help farmers solve this problem by choosing crops that take into account all factors, including planting season, soil and geographic location. Additionally, precision agriculture is gaining traction in emerging countries with an emphasis on site specific crop management and integration using modern agricultural techniques.

India is a predominantly agricultural country where the economy and the daily life of the people are strongly affected by agriculture. To increase agricultural productivity, it is recommended to grow crops according to soil, weather conditions, humidity, rainfall and other factors. This not only benefits farmers, but the country as a whole, and lowers food prices. This study demonstrates the use of machine learning techniques such as Random Forest and Decision Trees to predict which crops will perform best on a data set.

II. RELATED WORK

The most important actions to improve agriculture include the integration of technical knowledge and innovation to make agriculture more efficient and easier for farmers by using all ML techniques to predict ideal crops. This area discusses the benefits of various approaches including artificial neural networks, fuzzy networks, and data mining methods. Adding real time datasets to all of this is another issue.

One of the early works created a website dedicated to assessing the impact of weather conditions on agricultural

productivity in designated areas of Madhya Pradesh. Regions were selected according to crop area. Based on these criteria, the top five districts with the largest agricultural area were selected. Crop selection for that study was based on the dominant crops in the selected areas. The crops chosen were corn, soybeans, wheat and rice, and the yield of each was calculated for a continuous knowledge period of 20 years. For the selected crops, the models built were between 76% and 90% accurate, with an average accuracy of 82%.

Another important study assesses soil quality and predicts crop yields along with appropriate fertiliser recommendations. Model inputs include pH values and locations provided by the user. To predict the weather and temperature of the current location, an API is used. The system compares the results of supervised and unsupervised machine learning methods.

This is featured in classifiers for crop yield prediction using a greedy approach. It has been proven that using features in decision tree classifiers can improve performance. The proposed ensemble model has been shown to incorporate the effects of multiple models, often outperforming a single model. To predict agricultural yields, the random classification of forest sets uses several decision tree models. A ratio of 67% to 33% is used to divide the data into two groups, such as training data and test data, from which the mean and standard deviation are calculated. For the most accurate results, the method also groups related crops together.

Many ML methods have been used, and many efforts have been made in the field of agriculture. Increasing agricultural production and delivering it to consumers at the best price and with the best quality is the biggest problem in agriculture. It has also been noted that at least 50% of farm food is wasted

and never eaten. The proposed method provides suggestions for reducing wastage of agricultural products.

A recent paper presents a crop yield prediction model using the KNN algorithm and clustering. KNN clustering has been shown to perform better than SVM or regression. Using the latest regression techniques such as ENet, Lasso and Kernel Ridge algorithms, it can predict crop yield for a particular year. Using stack regression improves the accuracy of the algorithm.

Using the Pandas analysis tool, the historical dataset is filtered to extract the Maharashtra dataset. A crop yield prediction model is built using a multi layer perceptron neural network and tuned biases, weights, and Adam optimizers to improve accuracy. The proposed model predicts crop yield using an ANN with a three layer neural network. A crop yield forecasting system is built using supervised learning methods. Associations between many historical traits have been established and have allowed the system to increase crop yields.

Rainfall and temperature impact crop yields. To improve accuracy, recurrent neural network (RNN) and long short term memory (LSTM) algorithms are applied to this time series data. Using historical data, the ARMA (Autoregressive Moving Average), SARIMA (Seasonal Autoregressive Integrated Moving Average) and ARMAX (ARMA plus exogenous variables) methods are used to predict temperature and precipitation. In a crop yield forecasting system using fuzzy logic, the best model in the cluster is used.

Table I below is the table of comparative study of the IEEE papers which we have referred to regarding this recommendation system.

TABLE I
COMPARATIVE STUDY OF IEEE PAPERS

Sr.	Authors	Focus of the Paper	Key Points in the Coverage	Techniques Used	Parameters / Dataset	Research Gaps / Limitations
1	S. Pudumular, J. Nisha, C. Kavya, R. H. Rajashree	Develop a recommendation system through KNN and RF	Precision agriculture, recommendation system, ensemble model, KNN	KNN, Random Forest, Naive Bayes	Depth, texture, pH, soil permeability, water holding capacity	Less parameters and small dataset
2	Shilpa Pande, Dr. Prem, Anmol, B. R. Aishwarya, Karuna Rohilla, Kumar Shaurya	Predict crop yield for a user selected crop through RF and various ML algorithms	Crop yield prediction, ANN, KNN, Multivariate Linear Regression, fertilizer	SVM, RF, KNN, MLR, ANN	Soil, average rainfall, temperature	Fewer parameters; analysis of previous datasets not done
3	Nidhi Kulkarni, Dr. G. N. Srinivasan, Dr. B. M. Sagar, Dr. K. N. Cauvery	Classify input soil dataset into recommendable crop type (Kharif and Rabi)	Ensemble, majority voting, Naive Bayes, soil, crop recommendation	Ensemble learning, Linear SVM, Naive Bayes	Average rainfall, surface temperature, sowing season, NPK content of soil	Small dataset
4	Miftahul Jannat Mokarrama, Mohammad Shamsul Arefin	Recommendation system (RSF) for farmers that suggests the most suitable crops to produce in different areas	Crop recommendation, recommender system, similarity calculation	Social Pertinent Trust Walker (SPTW), CF based inference method	Location Detection Module, Data Analysis, Storage Module, Crop Production Rate Database	Less parameters and small dataset

Sr.	Authors	Focus of the Paper	Key Points in the Coverage	Techniques Used	Parameters / Dataset	Research Gaps / Limitations
5	Priyadharshini A., Aayush Kumar, Swapneel Chakraborty	Develop a recommendation system using Machine Learning	Profit analysis, crop recommender, crop sustainability predictor	CHAID, K Nearest Neighbor, Naive Bayes, Random Tree	Data analysis, data preprocessing, soil characteristics, environmental characteristics	Small dataset
6	Samarth Arlikar, K. A. Shinde, Shubham Kothavade	Predict crop yield based on climatic parameters, previous yield, and soil attributes	Machine learning and KNN	KNN	District name, season name, crop name; also predicts average cultivation and production obtained	Could also consider the economic aspect of farmers to recommend the most profitable crop
7	Avinash Kumar, Chittaranjan Pradhan	Crop recommendation system with pest control technique	SVM, pest control techniques, DT	DT, Logistic Regression, SVM	Types of soil, average temperature, pH, rainfall	Model can be developed further with soil attributes and a larger dataset
8	Zeal Doshi, Subham Nadkarni, Nipa Shah	Crop recommendation system using ML algorithms	ML algorithms, soil parameters, neural network	KNN, DT, RF, Neural Network	Rainfall, temperature, location, precipitation, pH, thickness	Model can be further improved to predict crop rotations to maximize yield
9	J. Madhuri, M. Indiramma	Develop a crop recommendation system using ANN	Climate, soil characters, ANN	ANN	Min/Max temperature, sunshine hours, soil texture, slope, depth, mean temperature, soil drainage	Works better only on a larger dataset
10	Avinash Devare, Rohit Kumar, Pooja Shinde	Maximize crop yield using ML algorithms	Precision agriculture, ANN, RF, DT	SVM, ANN, RF, Naive Bayes	pH, soil nutrients, rainfall, temperature, location	Dataset can be large with many attributes

III. METHODOLOGY

Despite numerous new solutions being put out, there are still unresolved issues with developing a user friendly website for crop advice. The proposed remedy seeks to address these drawbacks by creating a website that performs comparative analysis on the input crop and, based on the accuracy, recommends the crop. The system is a user friendly application that takes into account the factors that directly influence farming, such as rainfall, temperature, soil type, etc. The major goal is to increase the range of crops that can be cultivated throughout the season. By minimising the challenges farmers confront when selecting a crop and maximising productivity, the suggested system would ultimately assist in lowering suicide rates.

With the supplied data sets, the proposed model forecasts crop yield. By raising yields and maximising resource use, integrating agriculture and ML will help the agriculture sector go further. The most important factors in predicting present performance are historical data. Historical data is gathered from a number of trustworthy sources, including data.gov.in and kaggle.com. Some databases with state and district details include the soil type as an attribute. The retrieved soil type column is combined with the primary data set. In a similar manner, the major data sets for the particular location are supplemented with temperature and average rainfall from a different dataset. The data sets have been prepared and cleansed. New mean values are used to replace

the null values. Before running the algorithms, the categorical attributes are translated into labels.

A. System Architecture

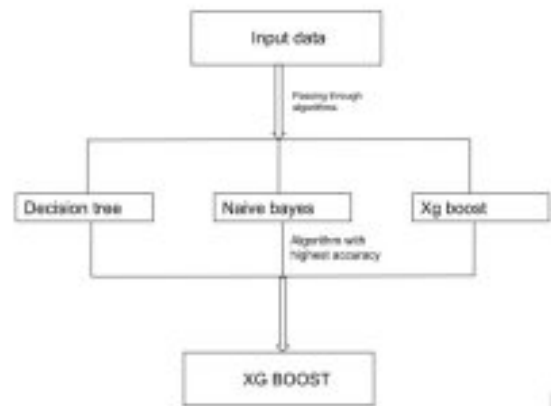


Fig. 1. System Architecture

Fig. 1 is the system architecture of the proposed algorithm. Algorithms are selected on the basis of the accuracy of each algorithm. In this model, the data was passed through 3 various algorithms and the algorithm having highest accuracy was selected.

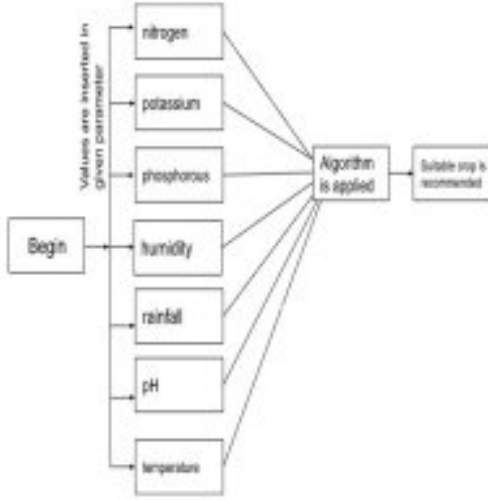


Fig. 2. Block Diagram

Fig. 2 is the system architecture of the process of crop recommendation, in which the farmer has to input various soil parameters given, and once the values are uploaded, the proposed algorithm is applied on the parameters. On the basis of the results a suitable crop is recommended to the farmers.

B. Algorithms Used

The algorithms which we have used for the implementation of this crop recommendation system are as follows.

1) Naive Bayes:

Classification problems are solved using the Naive Bayes supervised learning technique, which is based on the Bayes theorem. It primarily uses a huge training set for text categorization. The Naive Bayes Classifier, one of the simplest and most effective classification algorithms, facilitates the creation of quick machine learning models that are capable of making precise predictions. It provides predictions based on the likelihood that an object will occur because it is a probabilistic classifier. A few applications for Naive Bayes algorithms include spam filtration, sentiment analysis, and article classification.

2) Decision Tree:

Decision Tree is a member of the supervised learning algorithm family. Unlike other supervised learning techniques, the decision tree technique can handle classification and regression problems. A Decision Tree is used to create a training model that may be used to predict the class or value of the target variable by teaching simple choice rules generated from prior data. In decision trees, we start at the root when predicting the class label of a record. We compare the values of the attribute on the record with those of the root attribute. We jump to the node based on the comparison and follow the branch connected to that value.

3) XGBoost:

A distributed gradient boosting toolkit called XGBoost has been enhanced for speed and scalability in the training of machine learning models. Using the ensemble learning method, a number of weak models' predictions are pooled to produce a stronger forecast. Because it can handle massive datasets and excel at the forefront of many machine learning tasks, including classification and regression, Extreme Gradient Boosting, or XGBoost, is one of the most well known and widely used machine learning algorithms.

IV. RESULT ANALYSIS

The proposed system was implemented using the three selected algorithms and evaluated on the prepared crop dataset. The features in the dataset include nitrogen, potassium, phosphorus, humidity, rainfall, pH, and temperature, with the target being the recommended crop label. Fig. 3 illustrates the distribution of humidity across different crop labels, where each color in the scatter corresponds to a particular crop class.

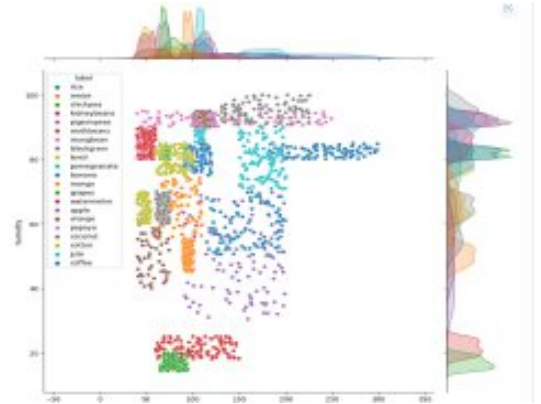


Fig. 3. Feature Distribution by Crop Label

To examine relationships between the seven input features, a pair plot was generated, as shown in Fig. 4. Each subplot in the matrix is a scatter of one feature against another, with the diagonal showing the per feature distribution. The plot helps in identifying clusters of crops that share similar nutrient and climate profiles.

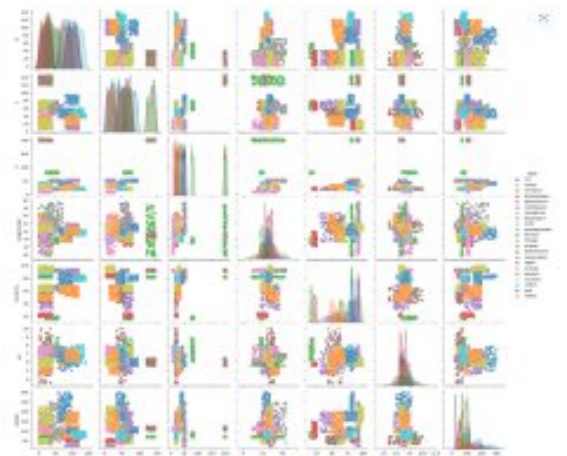


Fig. 4. Pair Plot of Input Features

The accuracy of each algorithm was compared after training and testing. As shown in Fig. 5, Decision Tree achieved approximately 90% accuracy, Naive Bayes approximately 99%, and XGBoost the highest accuracy at slightly above 99%. XGBoost was therefore selected as the final algorithm for the recommendation system.

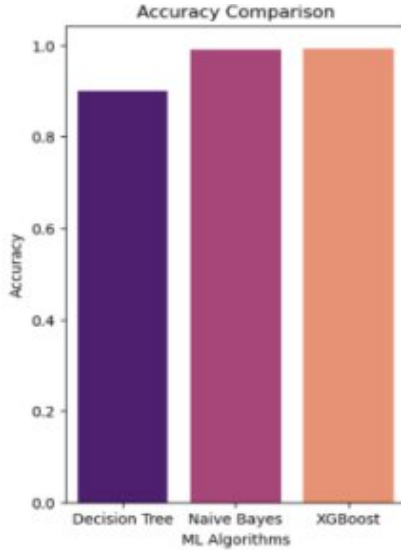


Fig. 5. Accuracy Comparison of ML Algorithms

Fig. 6 shows the correlation heatmap between the different input features. The matrix indicates that phosphorus and potassium are strongly correlated, while most other feature pairs are weakly correlated. This supports the feature independence assumption used by Naive Bayes and contributes to its strong performance on this dataset.

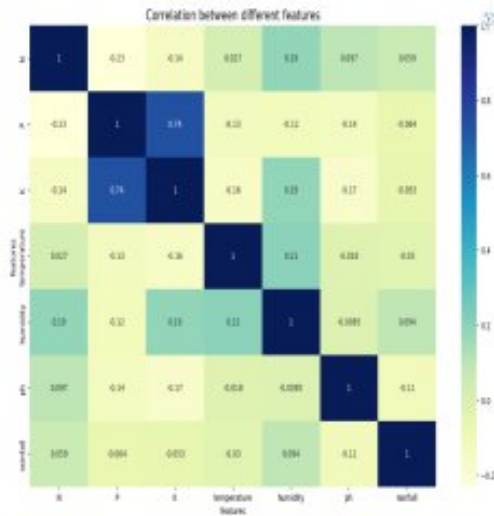


Fig. 6. Correlation Between Different Features

Above is the comparative analysis of the accuracy of the algorithms we have performed in this recommendation system.

V. CONCLUSION

The proposed method helps farmers choose the best crops by providing information that typical farmers do not monitor, thus reducing the likelihood of crop failure and increasing yields. It also protects them from losses. Millions of farmers across the country will in the future be able to access advice on planting crops through a combination of web interface and mobile app.

Future work will focus on updating the dataset from time to time to produce accurate predictions, and the process could be automated. Another feature that needs to be implemented is providing the right type of fertiliser for a given crop and location. To carry out this thorough study of available fertilisers and their relationship to soil and climate, must be done. Available statistical data should be analysed.

Thus, the work proposed makes it possible to help farmers make precise choices of crops to plant. This creates an exponential increase in crop productivity which, in turn, boosts the country's economy.

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